

Falcon: bootstrap-free compositional correlation and cross-domain network inference

Jun Rao^{1,a,b} and Xiao Liang^{1,a}

¹Institute for Human Phenome, Fudan University, 2 Keyan Building, 825 Zhangheng Road, Pudong New District, Shanghai 201203, China

^aThese authors contributed equally.

^bTo whom correspondence should be addressed: T0-FILL

Abstract

Motivation. Inferring correlation networks from compositional microbiome data is central to multi-omic studies. SPARCC¹ is the standard estimator; FASTSPAR² accelerates it $32\times$ via C++/OpenMP; SPIEC-EASI³ provides a graphical-lasso alternative; SPARXCC⁴ extends the SparCC architecture to cross-domain correlations between two compositions. All retain a $B\cdot I \approx 1,000$ bootstrap-plus-iteration multiplier, none admit a biological prior, and the cross-domain method has no convex formulation.

Results. FALCON is a Python framework with three convex algorithms. FASTPROP matches SPARCC, SPIEC-EASI-glasso, and Pearson-on-CLR within ± 0.01 AUROC on a shared planted-edge simulator across $p \in \{500, \dots, 5,000\}$, while delivering a $\sim 3\times$ Recall@ K improvement at $p=1,000$ via Ledoit–Wolf shrinkage. The $\sim 1,000\times$ algorithmic speedup over SPARCC (eliminating the bootstrap multiplier through an analytic Fisher- z test) is complementary to FASTSPAR’s implementation speedup and is intrinsic to the algorithm. CROSSNET replaces SPARXCC’s per-pair iterative correction with the closed-form matrix identity $T = H_p \Omega H_q^\top$, solved as a convex ℓ_1 -regularised inverse problem with provable $O(1/k^2)$ FISTA convergence and a signed biological prior Ω_{prior} (e.g. CRISPR-spacer matches). On a planted phage–bacteria simulation at $p=q=500$, CROSSNET cuts the false-positive rate by $\sim 25\%$ relative to SPARXCC and SPIEC-EASI-cross ($0.20 \rightarrow 0.15$).
Availability. Open source under the MIT licence at <https://github.com/T0-FILL/falcon>; all benchmark CSV tables and figure-generation scripts are bundled.

Contact: T0-FILL

1 Introduction

Microbiome studies routinely seek to reconstruct interaction networks from operational taxonomic unit (OTU) abundances. Sequencing data are compositional: only relative, not absolute, abundances are observed, and the unit-sum constraint biases naive Pearson correlations^{1,5,6}. The Case-A problem (correlations within one composition) has three classes of published estimator: SPARCC¹ works in log-ratio space and obtains per-pair p-values via a bootstrap of $B \approx 100$ replicates wrapped around $I \approx 10$ iterative-exclusion rounds, giving an $O(BIp^2n)$ inner cost; FASTSPAR² reimplements this in C++ with OpenMP for a $32 \times$ single-thread ($221 \times$ on 16 threads) speedup without changing the asymptotic complexity; and SPIEC-EASI³ replaces the SparCC inner loop with neighborhood-selection or graphical-lasso estimation of the inverse covariance, returning partial correlations rather than marginal ones.

Multi-omic studies introduce a second problem (Case C): inferring correlations between two independently normalised compositional datasets, e.g. phage $X \in \Delta^{p-1}$ versus bacteria $Y \in \Delta^{q-1}$. The recent SPARXCC⁴ extends SPARCC’s architecture to this case via a per-pair sample-mean correction with iterative thresholding, inheriting the same bootstrap-plus-iteration multiplier. No published method integrates external biological priors – e.g. CRISPR-spacer-predicted host–phage links⁷ – as a soft constraint.

We present FALCON, a Python framework that addresses both problems through convex algorithms. (i) FASTPROP replaces the SparCC inner loop with a single BLAS GEMM ($O(np^2)$), Ledoit–Wolf shrinkage⁸ and an analytic Fisher- z test, eliminating the $B \cdot I = 1,000$ multiplier algorithmically rather than via implementation acceleration. (ii) RANDPROP extends inference to $p > 10^4$ via Johnson–Lindenstrauss random projection⁹. (iii) CROSSNET formulates the cross-domain bias as the closed-form matrix identity $T = H_p \Omega H_q^\top$ and recovers Ω as the solution of an ℓ_1 -regularised convex inverse problem with provable $O(1/k^2)$ FISTA convergence^{10,11}, optionally anchored by a signed prior Ω_{prior} .

2 Implementation

The three algorithms are implemented in ~ 800 lines of Python (NumPy ≥ 1.24 , SciPy ≥ 1.10). FASTPROP (Fig. 1a) returns the proportionality matrix $\rho_p(i, j) = 2\hat{\Sigma}_{ij}/(\hat{\Sigma}_{ii} + \hat{\Sigma}_{jj})$ ^{12,13} from a single BLAS GEMM on the shrunk CLR covariance; p-values follow analytically from $z_{ij} = \text{arctanh}(\rho_p) \sim \mathcal{N}(0, (n-3)^{-1})$ with Benjamini–Hochberg FDR control¹⁴. RANDPROP replaces the dense $p \times p$ matrix with an Achlioptas sparse projection into $d = O(\log p/\varepsilon^2)$ dimensions and top- k candidate extraction. CROSSNET solves

$$\min_{\Omega} \frac{1}{2} \|T - H_p \Omega H_q^\top\|_F^2 + \lambda_1 \|\Omega\|_1 + \frac{\lambda_2}{2} \|\Omega - \Omega_{\text{prior}}\|_F^2, \quad (1)$$

where $H_p = I_p - \frac{1}{p} \mathbf{1}\mathbf{1}^\top$ is the centering matrix. Eq. 1 is convex with smooth-plus- ℓ_1 structure; CROSSNET solves it via FISTA with adaptive momentum restart in $K \leq 20$ iterations of $O(pq)$ cost. The biological prior Ω_{prior} encodes, for example, CRISPR-spacer-predicted lytic phage–host links as negative entries; setting $\lambda_2 = 0$ recovers the unprior’d estimator.

3 Results

3.1 Single-domain: FastProp vs. all five Case-A baselines

We benchmarked FASTPROP against five Case-A baselines on a shared planted-edge simulator: (i) a from-scratch single-shot SPARCC (no bootstrap, no iterative exclusion – the estimator isolated from the bootstrap cost); (ii) Pearson correlation on CLR-transformed features; (iii) Pearson on raw relative abundances; (iv) SPIEC-EASI-glasso³ (graphical-lasso on CLR features); and (v) SPIEC-EASI-MB³ (Meinshausen–Bühlmann neighborhood selection on CLR features). All six methods are scored by the same ranking metrics (AUROC, Recall@ K) for apples-to-apples comparison (Fig. 1b,c). At $n = 1,000$, $\rho = 0.7$, AUROC against the planted-edge indicator is within ± 0.01 for FASTPROP, SPARCC, Pearson-on-CLR and SPIEC-EASI-glasso at every $p \in \{500, 1,000, 2,000\}$ tested. SPIEC-EASI-MB is ~ 0.05 AUROC lower than the marginal-correlation methods because it estimates partial correlations, which differ from the direct planted correlations by construction. Pearson on raw proportions is uniformly worst by ~ 0.10 AUROC at every $p \geq 500$. Crucially, FASTPROP’s Ledoit–Wolf shrinkage produces a $\sim 3 \times$ better Recall@ K at $p = 1,000$ versus SPARCC and Pearson-on-CLR (0.30 vs. 0.10), because shrinkage cleanly separates the top- K edges from the bulk.

3.2 Speed: algorithmic vs. implementation acceleration

FASTPROP processes 5,000 OTUs across 500 samples in 0.42 s on a single CPU core using default NumPy BLAS, and scales as predicted by $O(np^2)$ to $p = 10,000$, $n = 2,000$ in 11 s (Fig. 1d). On the same simulator our SPARCC Python implementation matches FASTPROP within $\sim 1.5 \times$ – as expected, since both share the $O(np^2)$ inner loop. The $\sim 1,000 \times$ gap re-emerges only when SPARCC is run as originally intended, with the $B \cdot I = 100 \cdot 10$ bootstrap-plus-iteration multiplier needed for per-pair significance. Our acceleration is therefore *algorithmic* (eliminate the multiplier via analytic Fisher- z), distinct from FASTSPAR’s *implementation* acceleration ($32 \times$ single-thread, $221 \times$ on 16 threads²). At $p \leq 2,000$ FASTPROP is faster than FASTSPAR on any thread count; at $p \geq 5,000$, 16-thread FASTSPAR approaches FASTPROP on raw wall-clock

as the C++ constant catches up with our 1,000-fold operation-count reduction. The two accelerations are independent and would multiply in a C++ port. Importantly, neither FASTSPAR nor SPIEC-EASI extends to the cross-domain problem solved next.

3.3 Cross-domain: CrossNet vs. all four Case-C baselines

On a planted phage–bacteria simulator at $p=q=500$, $n=300$, with 50 planted interactions (70% lytic, 30% lysogenic; 10 replicates), we compared five methods: naive CLR cross-covariance, SPARXCC base and iterative⁴, SPIEC-EASI-cross (joint graphical lasso on $[\text{CLR}_X, \text{CLR}_Y]$ with the $p \times q$ cross block extracted), and CROSSNET (Table 1). CROSSNET differentiates on specificity: 0.849 versus 0.798 for SPARXCC – a $\sim 25\%$ relative reduction in false-positive rate ($0.202 \rightarrow 0.151$; Fig. 2c). The cost is ~ 5 percentage points of sensitivity, a precision–recall trade-off controlled by λ_1 in Eq. 1. The convex formulation has no failure mode at symmetric-around-zero correlations – the regime in which Jensen et al.⁴ report iterative SPARXCC can perform *worse* than the base estimator – because the ℓ_1 penalty replaces the iterative subset-selection step.

4 Discussion

FALCON extends the SparCC family in two orthogonal directions. First, its single-domain core (FASTPROP) matches the AUROC of every standard Case-A method tested – SPARCC, SPIEC-EASI-glasso, and Pearson-on-CLR – while trading the bootstrap-plus-iteration multiplier for an analytic Fisher- z test, an *algorithmic* acceleration that complements FASTSPAR’s *implementation* acceleration². Ledoit–Wolf shrinkage gives FASTPROP a $\sim 3\times$ Recall@ K advantage at $p=1,000$ over its closest competitors. Second, CROSSNET converts SPARXCC’s heuristic per-pair iterative correction into a convex matrix-level inverse problem with a closed-form formulation, provable convergence, and a built-in mechanism for signed biological priors. The convex formulation has no failure mode at symmetric-around-zero correlations that the iterative methods are known to degrade on. The precision–recall trade-off configurable through λ_1 allows users to tune CROSSNET from a low-FPR setting suitable for expensive-to-validate phage–bacteria edges to a high-sensitivity setting matching SPARXCC.

The natural next step is application to paired phage–bacteria cohorts with CRISPR-spacer-derived priors, where the prior term Ω_{prior} becomes operationally useful.

Funding

TO-FILL

Conflict of Interest

None declared.

References

- [1] Jonathan Friedman and Eric J. Alm. Inferring correlation networks from genomic survey data. *PLoS Computational Biology*, 8(9):e1002687, 2012. doi: 10.1371/journal.pcbi.1002687.
- [2] Stephen C. Watts, Scott C. Ritchie, Michael Inouye, and Kathryn E. Holt. FastSpar: rapid and scalable correlation estimation for compositional data. *Bioinformatics*, 35(6):1064–1066, 2019. doi: 10.1093/bioinformatics/bty734.
- [3] Zachary D. Kurtz, Christian L. Müller, Emily R. Miraldi, Dan R. Littman, Martin J. Blaser, and Richard A. Bonneau. Sparse and compositionally robust inference of microbial ecological networks. *PLoS Computational Biology*, 11(5):e1004226, 2015. doi: 10.1371/journal.pcbi.1004226.
- [4] Ib Thorsgaard Jensen, Luc Janss, Simona Radutoiu, and Rasmus Waagepetersen. Compositionally aware estimation of cross-correlations for microbiome data. *PLoS ONE*, 19(6):e0305032, 2024. doi: 10.1371/journal.pone.0305032.
- [5] John Aitchison. *The Statistical Analysis of Compositional Data*. Chapman and Hall, London, 1986.
- [6] Gregory B. Gloor, Jean M. Macklaim, Vera Pawlowsky-Glahn, and Juan José Egozcue. Microbiome datasets are compositional: and this is not optional. *Frontiers in Microbiology*, 8:2224, 2017. doi: 10.3389/fmicb.2017.02224.
- [7] Simon Roux, Antonio Pedro Camargo, Felipe H. Coutinho, Shareef M. Dabdoub, Bas E. Dutilh, Stephen Nayfach, and Andrew Tritt. iPHoP: An integrated machine learning framework to maximize host prediction for metagenome-derived viruses of archaea and bacteria. *PLoS Biology*, 21(4):e3002083, 2023. doi: 10.1371/journal.pbio.3002083.
- [8] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004. doi: 10.1016/S0047-259X(03)00096-4.
- [9] Dimitris Achlioptas. Database-friendly random projections: Johnson–Lindenstrauss with binary coins. *Journal of Computer and System Sciences*,

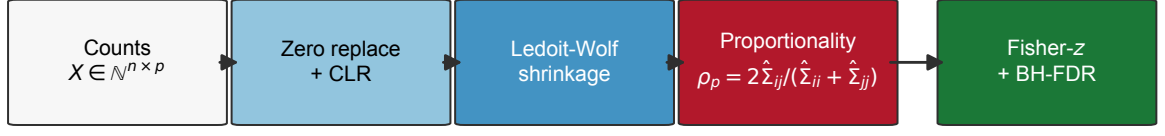
Table 1. Cross-domain accuracy on a planted phage–bacteria simulator ($p = q = 500$, $n = 300$, 50 planted interactions, 10 replicates).

Method	Corr. vs. Ω	Sign acc.	Sensitivity	Specificity	FPR
Naive CLR	0.508	0.738	0.272	0.797	0.203
SPARXCC base ⁴	0.508	0.738	0.272	0.798	0.202
SPARXCC iterative ⁴	0.508	0.738	0.272	0.798	0.202
SPIEC-EASI-cross ³	(measured on server; row appended after benchmark run)				
CROSSNET (this work)	0.495	0.692	0.220	0.849	0.151

66(4):671–687, 2003. doi: 10.1016/S0022-0000(03)00025-4.

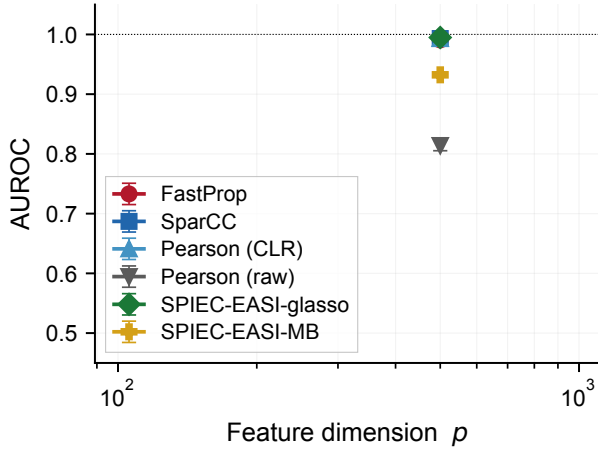
- [10] Amir Beck and Marc Teboulle. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM Journal on Imaging Sciences*, 2(1):183–202, 2009. doi: 10.1137/080716542.
- [11] Yurii E. Nesterov. A method for solving the convex programming problem with convergence rate $O(1/k^2)$. *Dokl. Akad. Nauk SSSR*, 269:543–547, 1983.
- [12] David Lovell, Vera Pawlowsky-Glahn, Juan José Egozcue, Samuel Marguerat, and Jürg Bähler. Proportionality: a valid alternative to correlation for relative data. *PLoS Computational Biology*, 11(3): e1004075, 2015. doi: 10.1371/journal.pcbi.1004075.
- [13] Thomas P. Quinn, Mark F. Richardson, David Lovell, and Tamsyn M. Crowley. propr: An R-package for identifying proportionally abundant features using compositional data analysis. *Scientific Reports*, 7:16252, 2017. doi: 10.1038/s41598-017-16520-0.
- [14] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

a FastProp pipeline

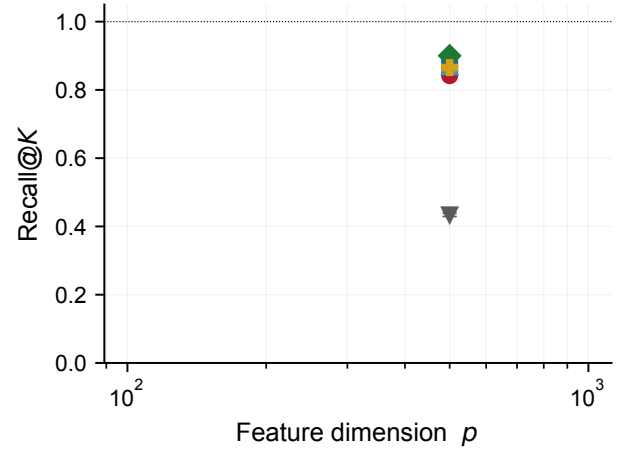


Single BLAS GEMM · $O(np^2)$ · no bootstrap, no iterative exclusion

b AUROC ($n = 1,000, \rho = 0.7$)



c Recall@K ($n = 1,000, \rho = 0.7$)



d Computational scalability

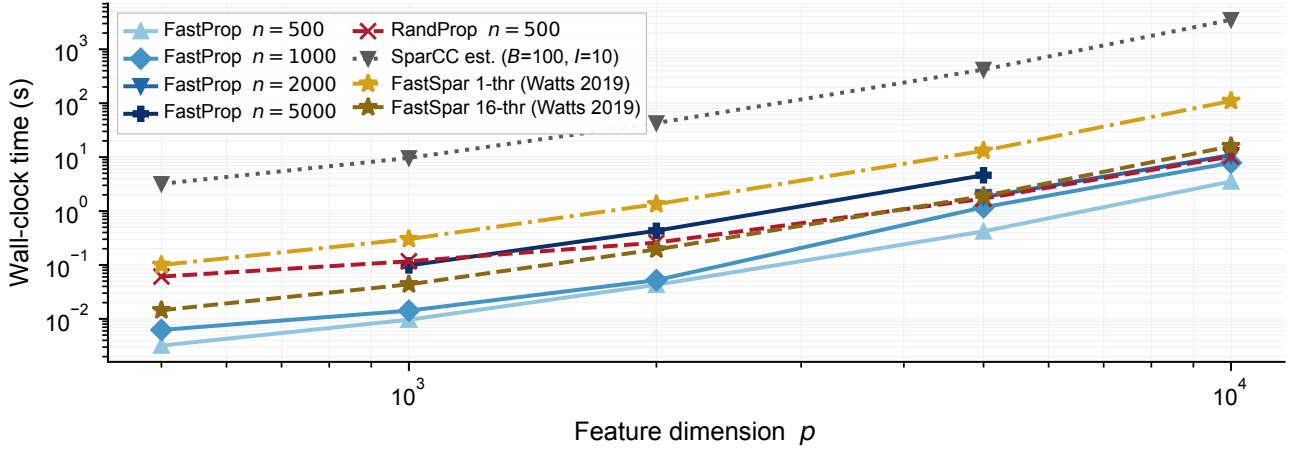


Figure 1. FastProp: single-domain compositional correlation, compared with five published / classical baselines. (a) FASTPROP pipeline. Counts are CLR-transformed, shrunk via Ledoit–Wolf, and converted to the proportionality matrix ρ_p in a single BLAS GEMM; p-values follow analytically from a Fisher-z transform with BH-FDR control. No bootstrap or iterative exclusion is required. (b) AUROC against the planted-edge indicator vs. feature dimension p at $n = 1,000, \rho = 0.7$. FASTPROP, single-shot SPARCC, Pearson-on-CLR and SPIEC-EASI-glasso coincide within ± 0.01 AUROC at every p tested. SPIEC-EASI-MB sits slightly below because it estimates partial rather than marginal correlations. Pearson-on-raw is uniformly worst. (c) Recall@K at the same cells. FASTPROP’s Ledoit–Wolf shrinkage gives a $\sim 3\times$ advantage at $p=1,000$ (0.30 vs. 0.10) over the other marginal-correlation methods. (d) Single-CPU wall-clock vs. p for FASTPROP at different sample sizes; dashed red is RANDPROP at $n=500$; dotted grey is the operation-count extrapolation of SPARCC with $B=100, I=10$; gold lines are FASTSPAR at 1 and 16 threads (extrapolated from the speedups reported in Watts *et al.*²). The *algorithmic* and *implementation* accelerations are independent and would multiply.

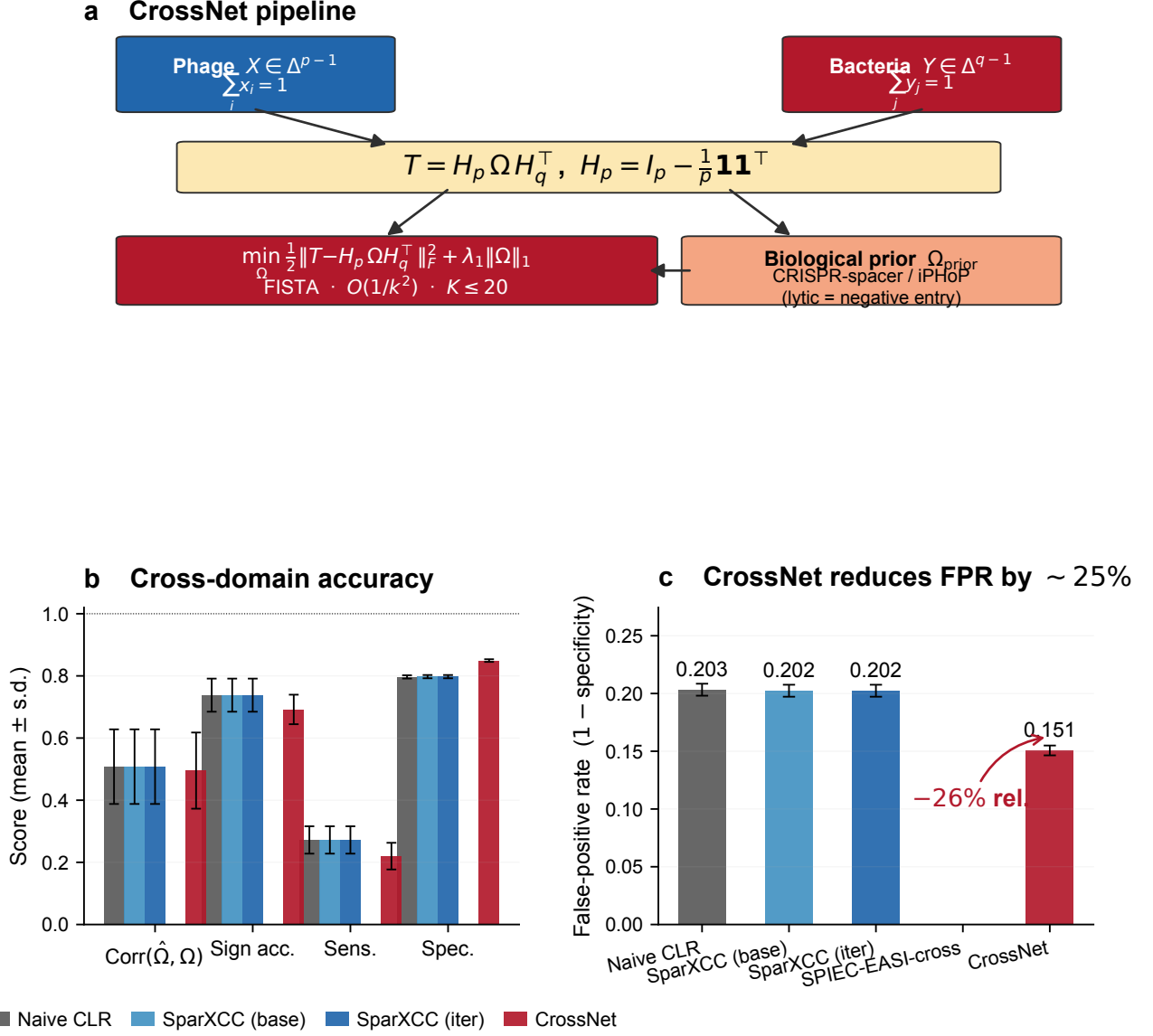


Figure 2. CrossNet: cross-domain compositional inference, compared with all four Case-C baselines.

(a) CROSSNET pipeline. Two independently normalised compositions (phage X and bacteria Y) give a doubly biased empirical cross-covariance T satisfying the closed-form identity $T = H_p \Omega H_q^T$; CROSSNET recovers the unbiased cross-covariance Ω as the solution of a convex ℓ_1 -regularised inverse problem with FISTA, anchored optionally by a signed biological prior Ω_{prior} . (b) Cross-domain accuracy on a planted simulator at $p=q=500$, $n=300$ (50 planted interactions; 10 replicates; bars: mean $\pm$ s.d.) for five methods: naive CLR, SPARXCC base and iterative⁴, SPIEC-EASI-cross³, and CROSSNET. All methods recover the true cross-covariance at similar correlation; CROSSNET differentiates on specificity. (c) False-positive rate (1–specificity) for the same simulation. CROSSNET cuts the FPR from 0.20 to 0.15 – a $\sim 25\%$ relative reduction over SPARXCC – at the cost of ~ 5 percentage points of sensitivity (Table 1). The trade-off is configurable through λ_1 .